

Human factors and ethics in technology

Experts in Teams (MA), 2024

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Motivation and agenda

- Human factors are essential for successful design and implementation of technological solutions (they constrain, hamper *and* aid construction and uptake)
- Not just a side-issue or something to be dealt with separately (or outsourced to specialists). Important to integrate a concern for human factors in the design process
 - Ethics; other values; value perceptions; typical reactions
(→ psychology and behavior science)
 - Integrating the concern for human factors in design processes
 - Methods for identifying and working with human factors

Advantages of taking into account/working with human factors

- Often easier to change user expectations or behavior than adapting the technology to existing behavior
- Striking success stories of cultivating preferences (e.g. iPad)
- Often less costly
- Adapting to existing behavior requires knowledge of it (e.g. waste sorting)
- Preventing later modifications of existing technology

Relevant human factors / values

- Ethics
- Moral psychology (*perceptions* and *ideas* about what is good or bad)
- Aesthetics
- Perceptions and experiences of convenience and inconvenience
- Influence on human wellbeing (affective responses etc.)
- Further psychological factors (determinants of human behavior)

1. Ethics – a very brief overview

Normative ethics: Theories that attempt to state the *general criteria* for right and wrong actions

Assumed to make our (: everyone's!) *implicit* ethics explicit (and systematic and coherent)

Two dominant theories:

Utilitarianism (popular form of consequentialism): An action is right if and only if it maximizes wellbeing

Deontology (“Ethics of duty”): An action is right if and only if conforms with basic ethic principles. The concern for maximizing wellbeing can be *overruled* by more fundamental concerns: not harming innocent people, respecting and supporting autonomy, dignity, property rights etc.

Ethics in practice

- The choice between utilitarianism and deontology is (often) not very important in practice. The two theories *converge* (support many of the same *specific* judgments)

In most cases, maximizing wellbeing *entails* not harming innocent people, supporting autonomy, avoiding deception etc. – and vice versa.

The concern for wellbeing is always *one* important concern (but perhaps among others)

→ Applied ethics: Assessing *specific practices* (domains): medicine, sports, journalism etc. – and *technology*, still more specifically e.g. *robotics*)

Approaches to *technology* ethics

- Classical applied ethics: Applying utilitarianism and/or deontology directly to cases of technology development (e.g. will the device lead to more wellbeing overall?) (“Top-down”)

Criticism: Ethical assessment should be more sensitive to the specific domain in question (e.g. technology; robotics)

- Bottom-up applied ethics (Beauchamp, 1984) (working back and forth between application and theory; allowing the theory to be twisted)
- “Ethics by design” (recommended by the EU) (“frontloaded”; ethics constrains the design process from the start)
- Ethics *in* design (“Values Sensitive Design”; integrating ethics in the (whole) design process)

EU ethical guidelines (“Ethics by design”) (for AI *and* robotics)*

There are six general *ethical principles* that any AI system must preserve and protect based on fundamental rights as enshrined in the Charter of Fundamental Rights of the European Union (EU Charter), and in relevant international human rights law:

1. **Respect for Human Agency:** human beings must be respected to make their own decisions and carry out their own actions. Respect for human agency encapsulates three more specific principles, which define fundamental human rights: **autonomy**, **dignity** and **freedom**.
2. **Privacy and Data governance:** people have the right to privacy and data protection and these should be respected at all times;
3. **Fairness:** people should be given equal rights and opportunities and should not be advantaged or disadvantaged undeservedly;
4. **Individual, Social and Environmental Well-being:** AI systems should contribute to, and not harm, individual, social and environmental wellbeing;
5. **Transparency:** the purpose, inputs and operations of AI programs should be knowable and understandable to its stakeholders;
6. **Accountability and Oversight:** humans should be able to understand, supervise and control the design and operation of AI based systems, and the actors involved in their development or operation should take responsibility for the way that these applications function and for the resulting consequences.

*https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ethics-by-design-and-ethics-of-use-approaches-for-artificial-intelligence_he_en.pdf

Limitations of the EU guidelines?

- The principles mention several *values* that are not further defined (e.g. dignity, privacy etc.). When is an advantage “undeserved”? What are “equal opportunities”? Who are the affected (: relevant group of “people”)?
 - The proof of the pudding is in the eating! What are the *specific* values and concerns that are relevant to / appropriate in the specific case? How should the principles and values be *interpreted*?
 - The need for integrating ethics in the design process/working back and forth between application (domain and stakeholders) and theory

The EU guidelines – elaborations, but also new questions (and an important observation)

- **Autonomy:** Respecting autonomy means allowing people to think for themselves, decide for themselves what is right and wrong, and choose how they want to live their lives. It is important to note that AI systems can restrict human autonomy without doing anything - simply by not catering for the full range of human variation in lifestyle, values, beliefs and other aspects of our lives which make us unique. Hence, such restrictions may arise without any malevolent intent, but solely as a result of a lack of understanding about peoples' lives, beliefs, values, and preferences. This is a particular problem with personalisation services which do not take into consideration varying cultural norms. In addition, personalisation services, even when taking into consideration varying cultural norms, might restrict the information and options provided. AI based systems should avoid restricting unreasonably individual decision-making context (see also below under "freedom").
- **Dignity:** Dignity entails that every human being possesses an intrinsic worth which should never be compromised. Hence, people should not be instrumentalized, objectified or dehumanized, but must be treated with respect at all times, including when using or being subjected to AI-based systems.
- **Freedom:** Respecting freedom requires that people are not constrained in taking those actions which they should be able to pursue as autonomous persons, such as freedom of movement, freedom of speech, freedom of access to information, and freedom of assembly. In addition, freedom requires the absence of constraints which undermine peoples' autonomy, such as coercion, deception, exploitation of vulnerabilities, and manipulation.

Indirect ethical consequences. Need for ethical *imagination*

”not catering for the full range of human variation in lifestyle, values, beliefs and other aspects of our lives which make us unique. Hence, ... restrictions [of autonomy] may arise without any malevolent intent, but solely as a result of a lack of understanding about peoples’ lives, beliefs, values, and preferences”.

- The importance of human factors!
- Risk of doing harm by overlooking interests of certain groups (e.g. people with disabilities, different body sizes, culturally conditioned preferences)

AbilityNet In partnership with **BT Group**

"66% of people who care for a loved one with dementia believe technology can enhance their quality of life"



Examples: Autonomy and transparency matters, but

Dementia: What if people cannot tell or act on what they prefer? Whose preferences (one's former self? Relatives? ("Relational autonomy"))

Transparency – or rather *trustworthiness*? How far is transparency *realistic*? Risk of *information overload* (or pseudo-transparency)



USEFUL TECH FOR DEMENTIA CARE

Smart Locks 	Smartwatches 	Automatic Pill Dispenser 
Emergency Call Buttons 	GPS Tracking 	Digital Clocks/Calendars 

© Marmage

Limitations of classical technology ethics

- Evaluation done before (“frontloading”) or after the design process.
- Difficult to determine in advance which ethical concerns will turn out to be relevant
- Costly and unrealistic to discard or substantially modify an otherwise successful technological device once it has been constructed
- Tendency to focus on “big” issues (large security risks, violation of human rights etc. Real-world ethical issues are often more mundane (not violation of rights or danger to human lives, but risks of causing inconvenience, disturbance, making people uncomfortable, worried etc.
- Tendency to focus one-sidedly on *risks* and *prohibitions*. Ethics can and should also be about making things *better*. (Technology can *strengthen* wellbeing, autonomy, dignity etc.)

2. Other human factors (especially in robotics)

- Large unrealized potential of automation and robot use
- Robotics technology still very far from perfect – huge construction challenges – but existing solutions are also not used to the full extent

Successful human-robot collaboration: Joint dependence on robot factors (RF) and human factors (HF)

Robots can behave in different ways. But a single behavior can also trigger different reactions and perceptions (in different users, or in different circumstances)

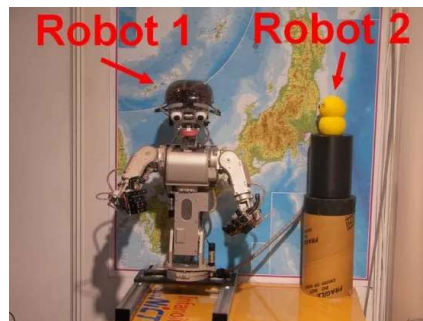
Other than ethical values



Aesthetics

(beautiful, ugly, harmonious, elegant, well-proportioned/lacking of proportion), unitary/coherent etc.)

- Aesthetics is ethically relevant: Large impact on *wellbeing*
- Aesthetics is practically relevant: Central to *acceptance*



Values and experiences in human-robot interaction (etc.)



- Trust (mediated (enhanced) by *reliability perception* (as is trust in humans). Distrust can stem solely from perceptions of novelty and strangeness, but is enhanced by unreliability perceptions (→ importance of not exposing users to premature, unreliable design)
- Proximity can foster acceptance (getting accustomed to the robot) but also anxiety and inconvenience (safety issues) (collision avoidance)
- Working with robots (or their mere presence) can cause *anxiety*, increased mental strain and perceived higher workload. Typical situation: The human as supervisor (monitoring; compensating for the unexpected). Difficult situations being left to humans to deal with; lack of perceived support or empathy). Being in the way without actually being in the way



Experiment 1

* Question:

- Can tool be embodied?

* Measures:

- Proprioceptive drift
- Questionnaire



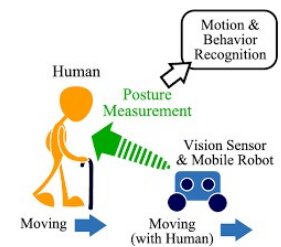
An ideal (?): Tool embodiment

When functioning optimally, *tools* are not perceived as external objects, but as (or like) the user's own body parts (integrated into the action flow)

Human factors for robots!

- Not just human perception of robots
- Robots can also need to "perceive" human behavior (decode human intentions; predict human behavior) (reacting to humans' behavior can be more effective (and considerate) than acting on their *commands*).

→ Need to know how humans tend to behave in a specific environment

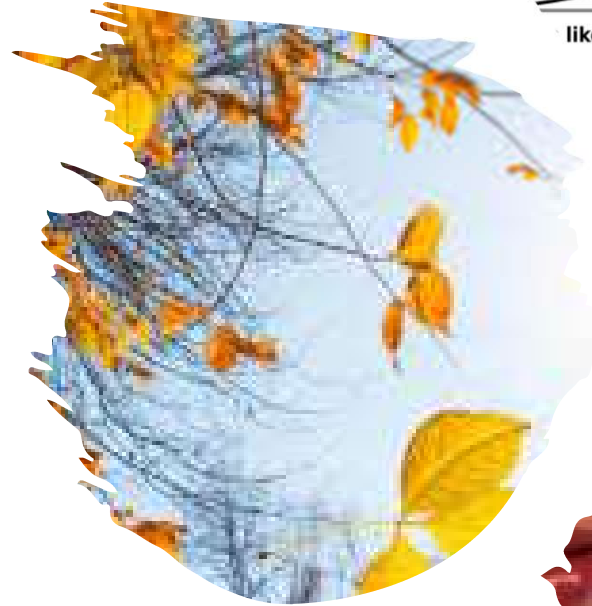
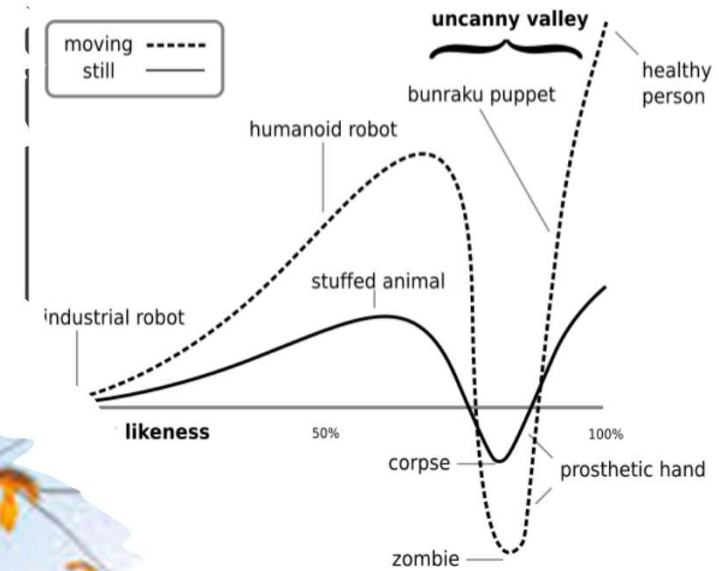


Turtlebot2 reasoning about a potential occluded human based on interactive behaviors of the observed humans.

Making it human – but not all too human?

Perceiving humans as humans: A combination (entanglement) of *necessity* and *freedom* (acting orderly and to some degree expectedly (or predictable – not *randomly*), but also not *automatic* or *completely* predictable (still displaying *autonomy*))

The uncanny valley: unsettling or eerie feeling caused by robots etc. that closely, but not completely resemble humans



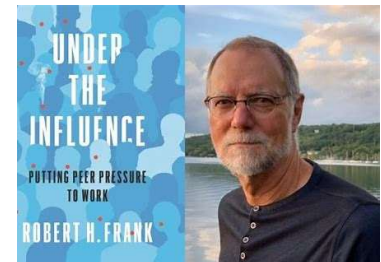
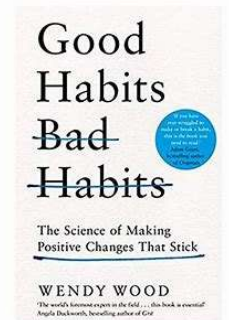
Design principles for avoiding the uncanny valley

- Avoid mixing degrees of realism (mixing human and non-humanlike elements) (compare a human with a wooden leg)
- Match appearance, behavior and ability (compare a human corpse)



Further psychological factors: Learning from behavioral science

- People are **motivated by experienced wellbeing and convenience** (and perceptions of potential immediate changes to their wellbeing)
- Behavior change requires **repetition, reward, and a supportive environment** (Wood, 2020)
- Social support, imitation and **peer pressure** (Frank, 2020) (the power of behavioral contagion; acceptance thresholds and acceptance cascades)



How to work with perception and values?

- 1) Identify stakeholders. Who will (or can) be affected by the device?
(Snowball sampling: Use known stakeholders to identify further stakeholders)
- 2) Elicit and record stakeholder perceptions
- 3) Balance and interpret stakeholder perceptions with ethical theory/ethical principles

Eliciting perceptions: Individual *interviews* / focus group *interviews*

Using *iteration*: Present still more specific solutions to stakeholders

Usually *semi-structured* interviews: Prepared questions, mentioning particular focus points (potential concerns) – but open to where the conversation leads)

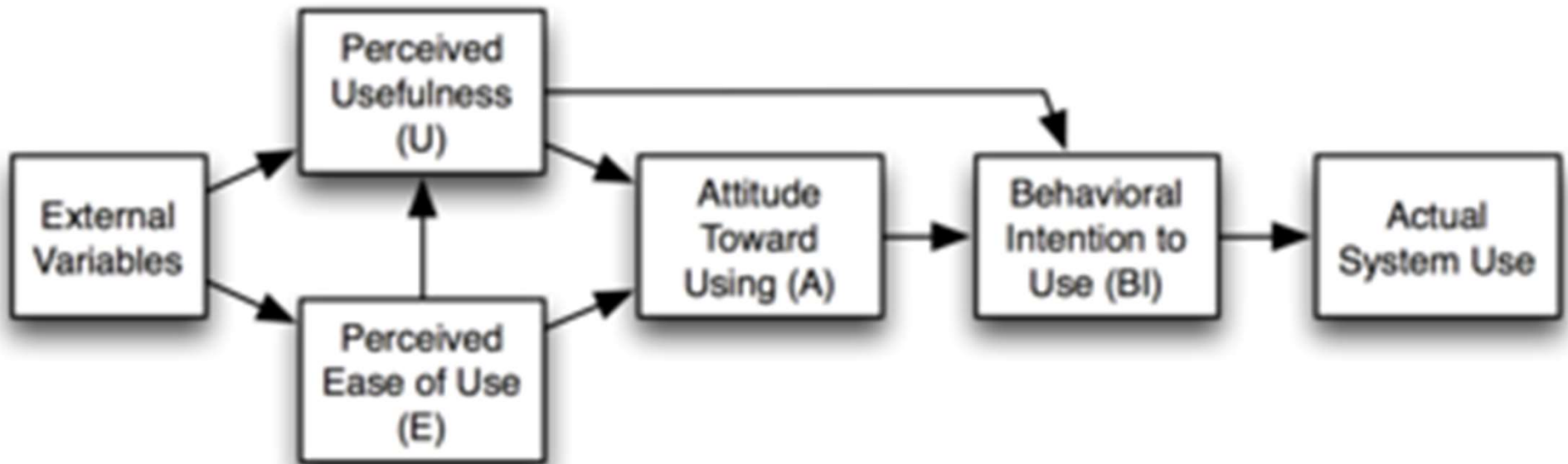
Balancing reported perceptions with other concerns

- Use reported perceptions and concerns as a rough guide
- But correct and complement them, based on
 - Ethical theory and ethical guidelines (stakeholders may be ignorant of, or just not report, important concerns)
 - Knowledge of potential biases or misapprehension (a fine balance: “unfounded worries” may have to be taken seriously; but some worries are not practically relevant)

Knowledge that humans are prone to *adapt* and modify their values

The strength of in-depth interviews and focus groups (as opposed to surveys): Possible to negotiate and refine (“calibrate”) answers

The strength of complementary *observations*: People don’t always notice their own reactions



Coda: The Technology Acceptance Model (Adams et al. 1992)

Useful, but ignores the impact of habituation and overestimates the impact of attitudes.
Also ignores more fundamental values (other than usefulness and ease of use)

References

Adams, D. A; Nelson, R. R.; Todd, P. A. (1992), "[Perceived usefulness, ease of use, and usage of information technology: A replication](#)", MIS Quarterly, **16** (2): 227–247, [doi:10.2307/249577](#), [JSTOR 249577](#), [S2CID 12996460](#)

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